

Answer Key — Pure Substances and Elements

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **cannot be broken into simpler substances** — elements cannot be broken down into simpler substances.
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- Q2** (a) **atom** — each element is made of identical atoms.
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- Q3** (b) **carbon, sulfur, oxygen** — carbon, sulfur and oxygen are non-metals.
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- Q4** (c) **metalloids** — elements between metals and non-metals are metalloids.
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- Q5** (a) **Both true, R explains A** — since water splits into two elements, it isn't an element — as R explains.
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- Q6** **molecule** — joined atoms form a molecule.
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- Q7** **particle** — a pure substance has one kind of particle.
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Short answer

- Q8** In science, a pure substance is made of only one type of particle and can't be split physically. Everyday 'pure' means unadulterated, but such a product may still be a mixture of several substances.
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- Q9** A pure substance that cannot be broken into simpler substances; metals: gold, iron (etc.); non-metals: carbon, sulfur, oxygen (any two each).
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- Q10** An atom is the smallest particle of an element; a molecule is two or more atoms joined together.
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Long / application

- Q11** All are elements (cannot be broken into simpler substances). Gold — metal; oxygen — non-metal; silicon — metalloid; sulfur — non-metal. An element is a pure substance made of one kind of atom.
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- Q12** To a scientist 'pure' means a single kind of particle, so ghee (many substances) would be a mixture, not pure. On the label, 'pure' means unadulterated — nothing cheaper or harmful has been added — which is the everyday meaning and acceptable for food.
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